Leverage Cycles over the Household Life Cycle: A Quantitative Model of Endogenous Leverage

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Introduction

Introduction

Empirical facts: In U.S. housing cycles,

$$\text{house prices} \uparrow \quad \text{leverage} \uparrow \quad \text{mortgage spreads} \downarrow .$$

Question: Can a model rationalize this joint behavior of housing prices, leverage, and mortgage spreads? Who ultimately bears the cost of a financial crisis?

Why leverage matters

Leverage measures debt relative to collateral value:

$$LTV = \frac{\text{outstanding debt}}{\text{value of collateral}}.$$

- It allows households to acquire housing without making large payments, but exposes borrowers to housing-price risk and lenders to default risk.
- In aggregate, leverage amplifies fluctuations in household wealth and housing prices.

Introduction

Benchmark in macro-housing models (Kiyotaki and Moore, 1997; Iacoviello, 2005; Iacoviello and Neri, 2010; Favilukis, Ludvigson and Van Nieuwerburgh, 2017):

$$b_t \leqslant \phi q_t h_t.$$

- Borrowing is modeled as a single riskless bond up to an exogenous fraction of collateral value.
- Default is ruled out.
- Kaplan, Mitman and Violante, 2020: with default, but mortgages are priced by risk-neutral deep-pocketed lenders

This paper: Endogenizes the pricing of leverage. Geanakoplos (1997), Geanakoplos and Zame (2014) and Fostel and Geanakoplos (2008, 2015).

- Households choose leverage from a menu of defaultable mortgage contracts.
- Each contract corresponds to a different LTV and is priced by lenders at a different interest rate.
- Hence, leverage is priced at the margin: higher LTVs carry higher default risk and higher mortgage rates.
- This allows for default risk, and actual default to arise in equilibrium,

Model

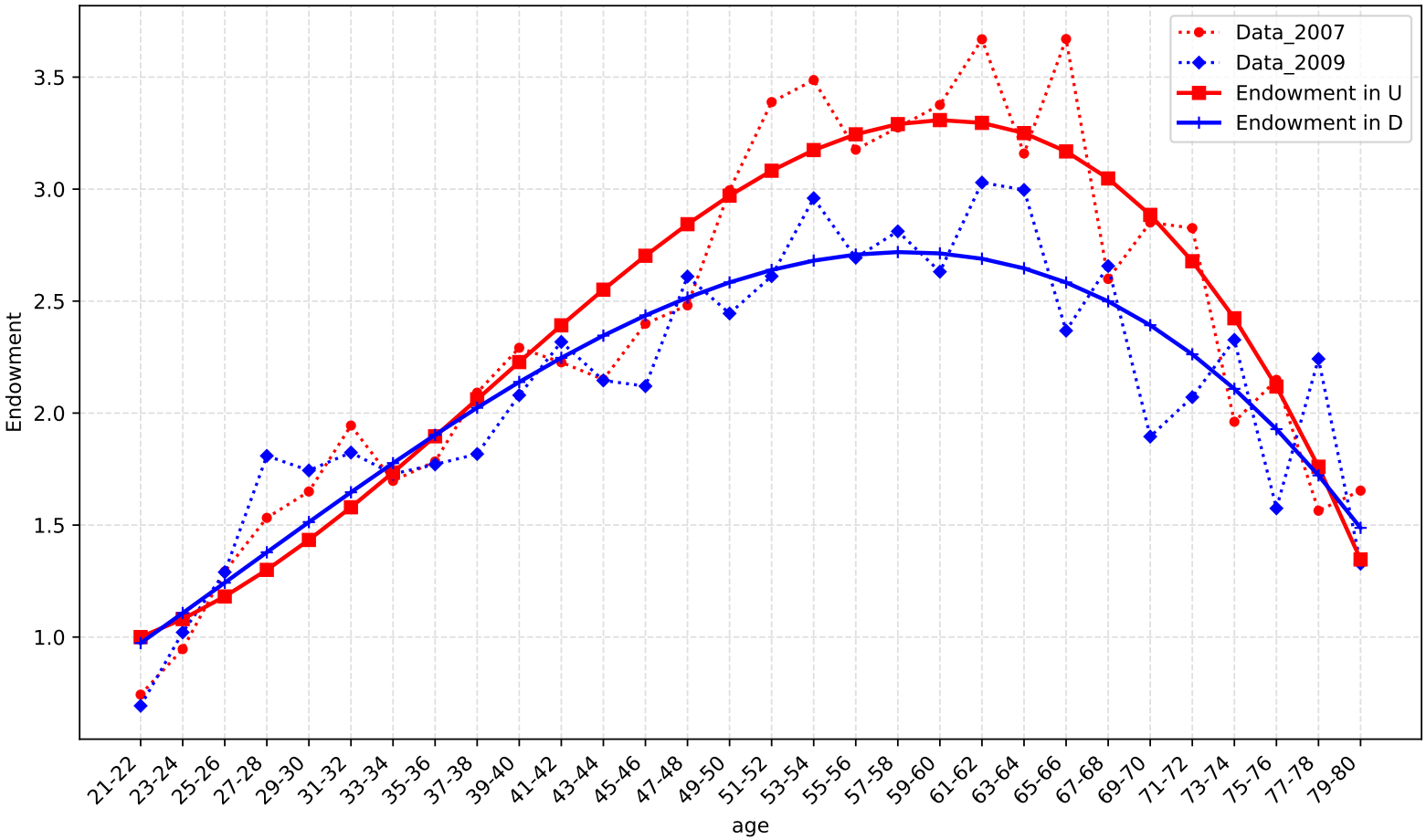
Model: Environment

- **Age**: $a \in \{1, \ldots, A\}$
  - No uncertainty about lifetime
  - No heterogeneity within cohorts
- **Uncertainty**
  - $z_t \in \{U, D\}$, Markov process
- **Commodities**

		Spot Price	Agg. Supply
Consumption (c_t^a)	perishable	1	$\bar{e}(z_t)$
House (h_t^a)	perfectly durable, divisible	q_t	H

- **Endowments**
  - $\bar{e}(z_t) = \sum_{a=1}^{A} e^a(z_t) = \sum_{a=1}^{A} e_t^a$

Endowment age profiles



Model: Preferences

- Period Utility

$$u^a(c,h) = \begin{cases} \dfrac{(c^{1-\alpha}h^{\alpha})^{1-\rho}}{1-\rho} & a \in \{1,...,A-1\} \\ \dfrac{c^{1-\rho}}{1-\rho} & a = A \end{cases}$$

- Lifetime Utility

$$U_t = E_t \sum_{a=1}^{A} \beta^{a-1} u^a(c_t^a, h_t^a)$$

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  - $\theta^a_{j,t} \in \mathbb{R}$
    - $\theta^a_{j,t} < 0$: short, borrow $|\pi_{j,t}\theta^a_{j,t}|$
    - $\theta^a_{j,t} > 0$: long, lend $|\pi_{j,t}\theta^a_{j,t}|$
  - Delivery: $\min\{j, q_{t+1}\}$
    - Non-recourse

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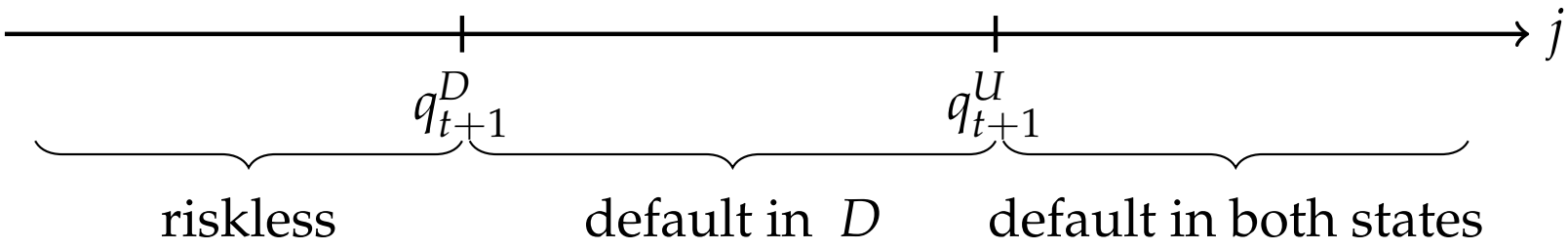
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  } Credit Surface: CS_t

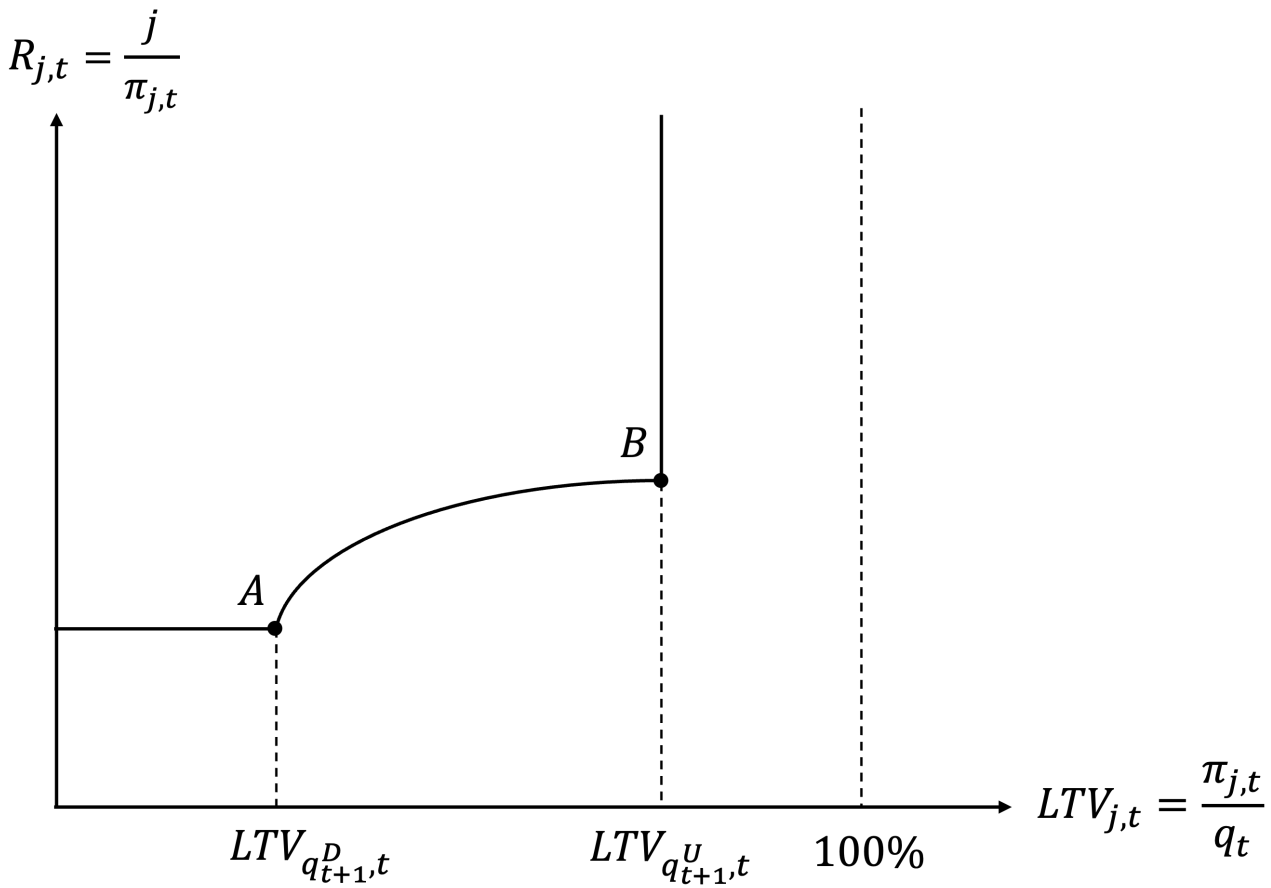
Credit Surface

At date t, two possible housing prices tomorrow: $q^D_{t+1} \leqslant q^U_{t+1}$:



Credit Surface (CS_t)

Following Fostel and Geanakoplos (2015)



Constraints

- **Budget Constraint**

$$c_t^a + q_t h_t^a + \int_{R_+} \theta_{j,t}^a \pi_{j,t} dj \leqslant e_t^a + q_t h_{t-1}^{a-1} + \underbrace{\int_{R_+} \theta_{j,t-1}^{a-1} \min\{j, q_t\} dj}_{\text{Total delivery of contracts}}$$

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$$\underbrace{\int_{j>q_t} q_t \, \theta_{j,t-1}^{a-1} dj}_{\text{Default: deliver } q_t} + \underbrace{\int_{j\leqslant q_t} j \, \theta_{j,t-1}^{a-1} dj}_{\text{Repay promised } j} = \longleftarrow$$

Constraints

- **Collateral Constraint**

$$\int_{R_+} \max\{-\theta^a_{j,t}, 0\} dj \leqslant h^a_t$$

- **No short-selling Constraint**

$$h^a_t \geqslant 0$$

Definition: Collateral Equilibrium

- Allocations $\{c_t^a, h_t^a, (\theta_{j,t}^a)_{\forall j \in J_t}\}\ \forall t,\ a$
- Prices $\{q_t, (\pi_{j,t})_{\forall j \in J_t}\}\ \forall t$

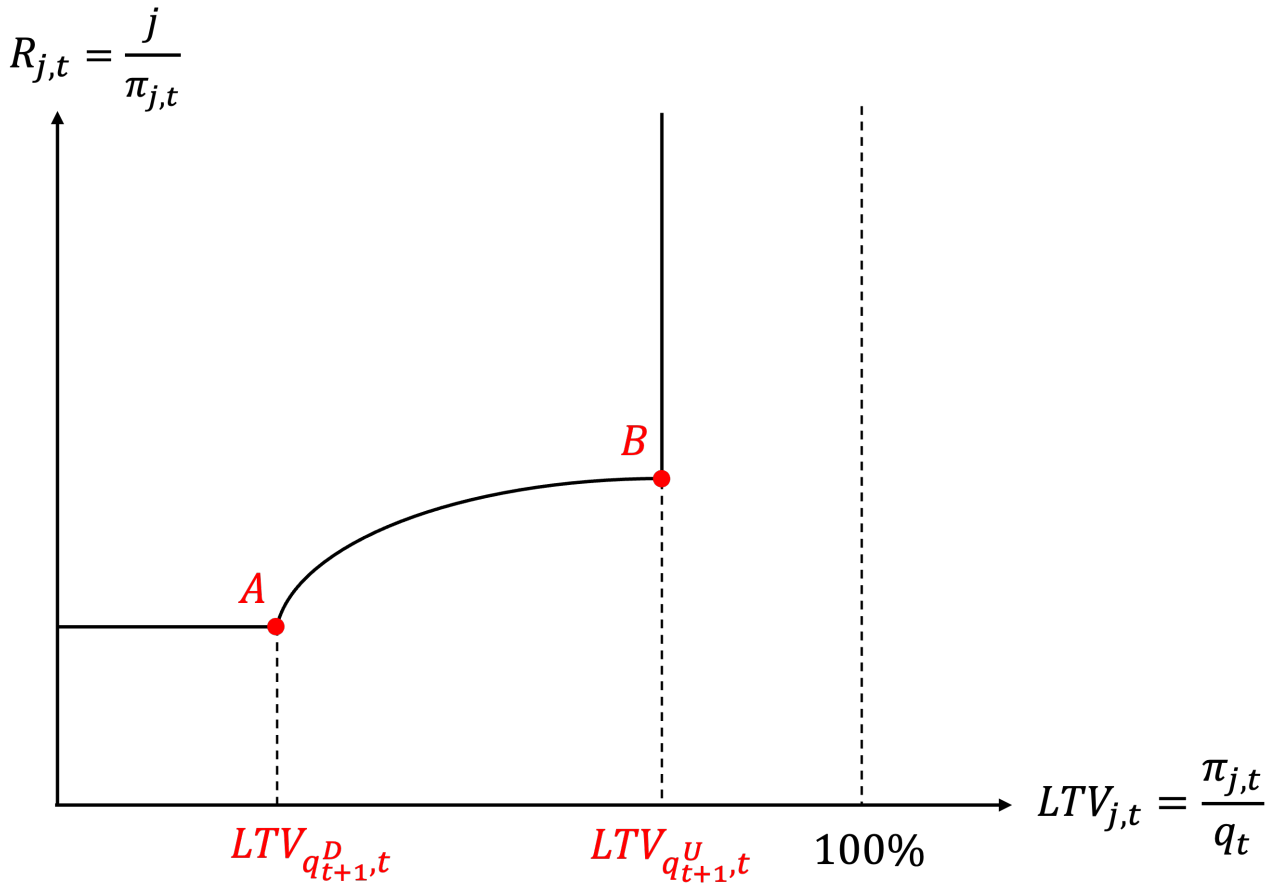
s.t.

- Households maximize life-time utility subject to the constraints, taking prices as given
- All markets clear

Results

Equilibrium

- 2 contracts traded in each period: Riskless: q^D_{t+1} and Risky: q^U_{t+1}



Equilibrium

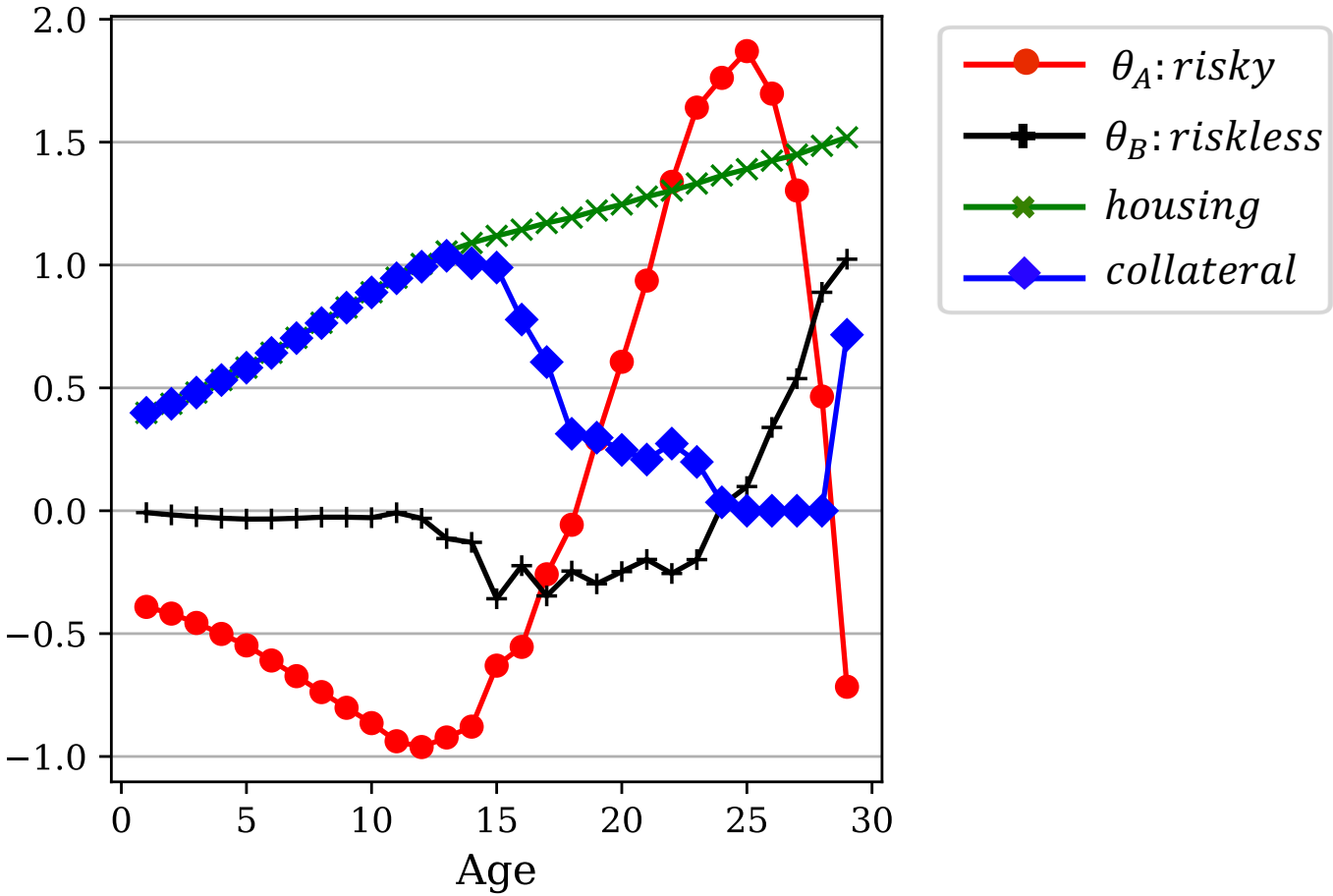


Figure: Portfolio Holdings (Mean)

Constrained borrowers

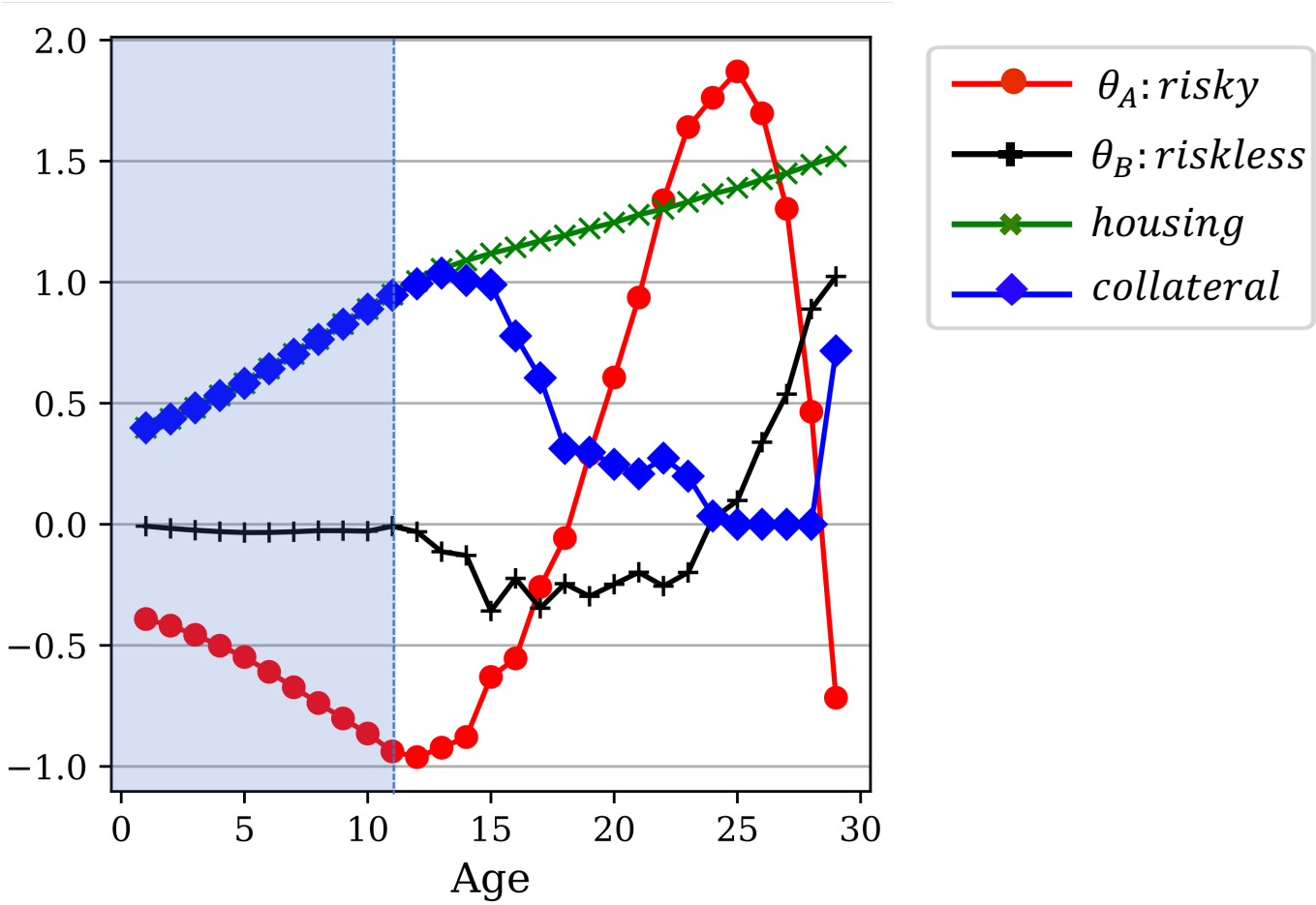


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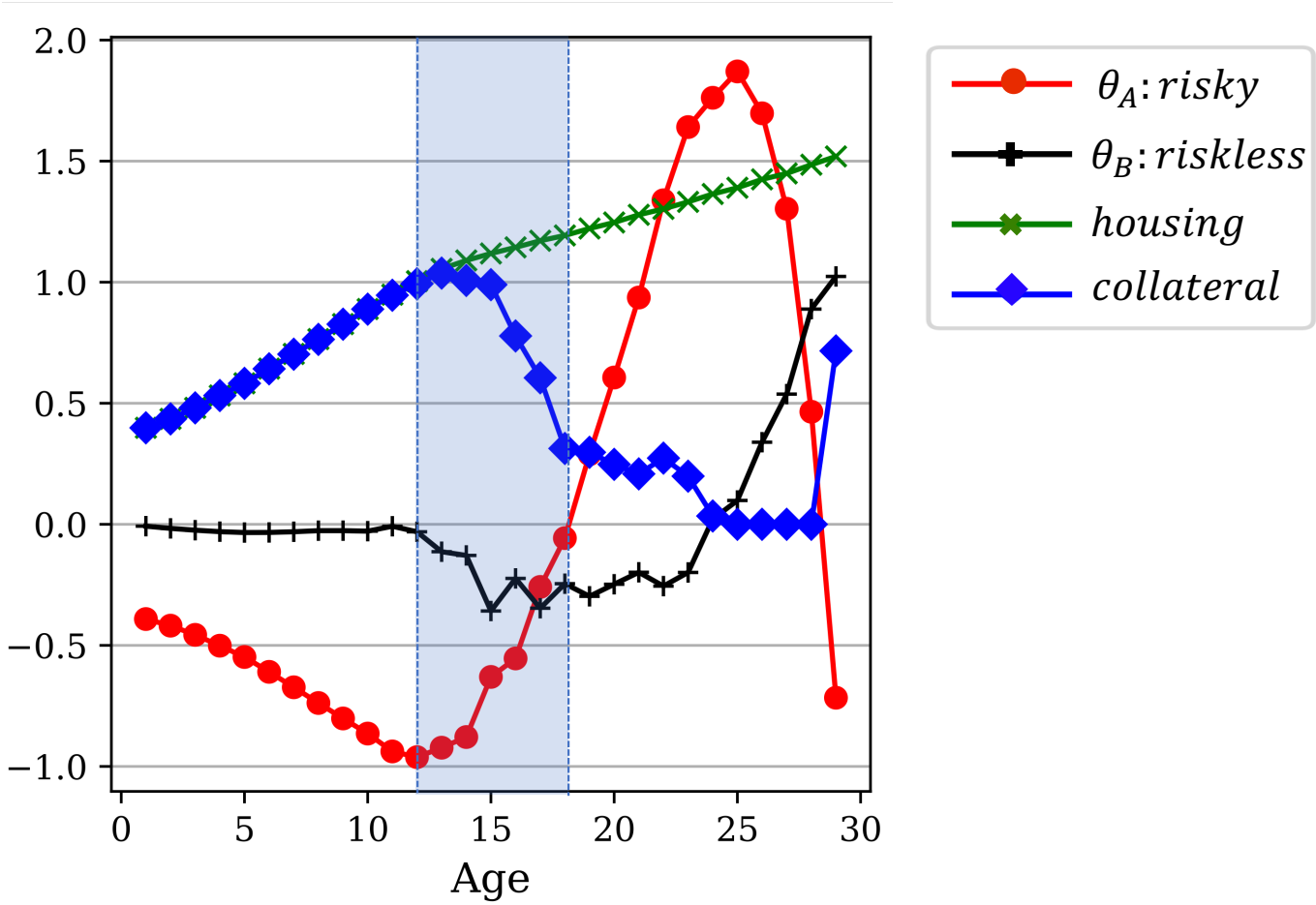


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Long Risky, Short Riskless

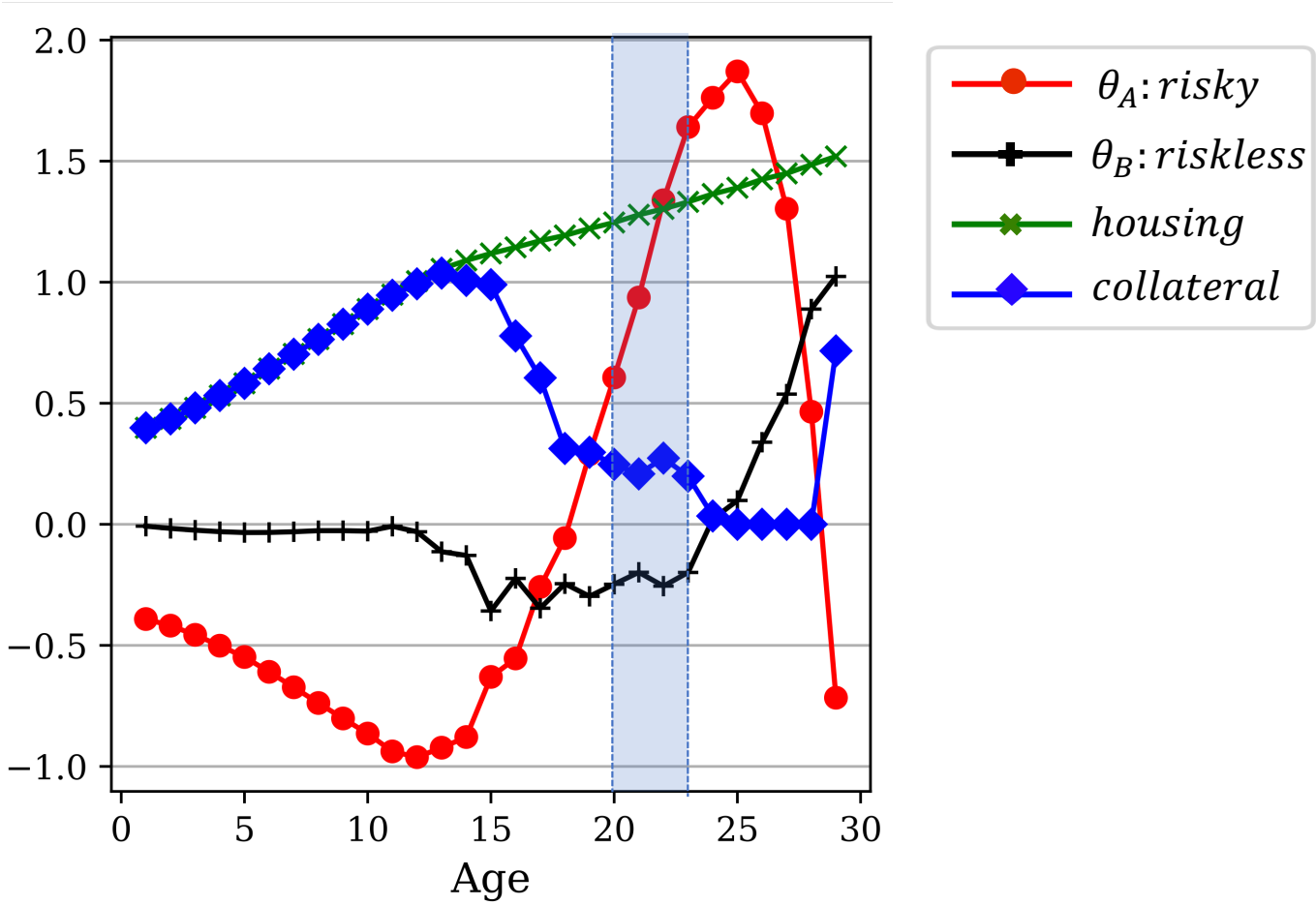


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Lenders

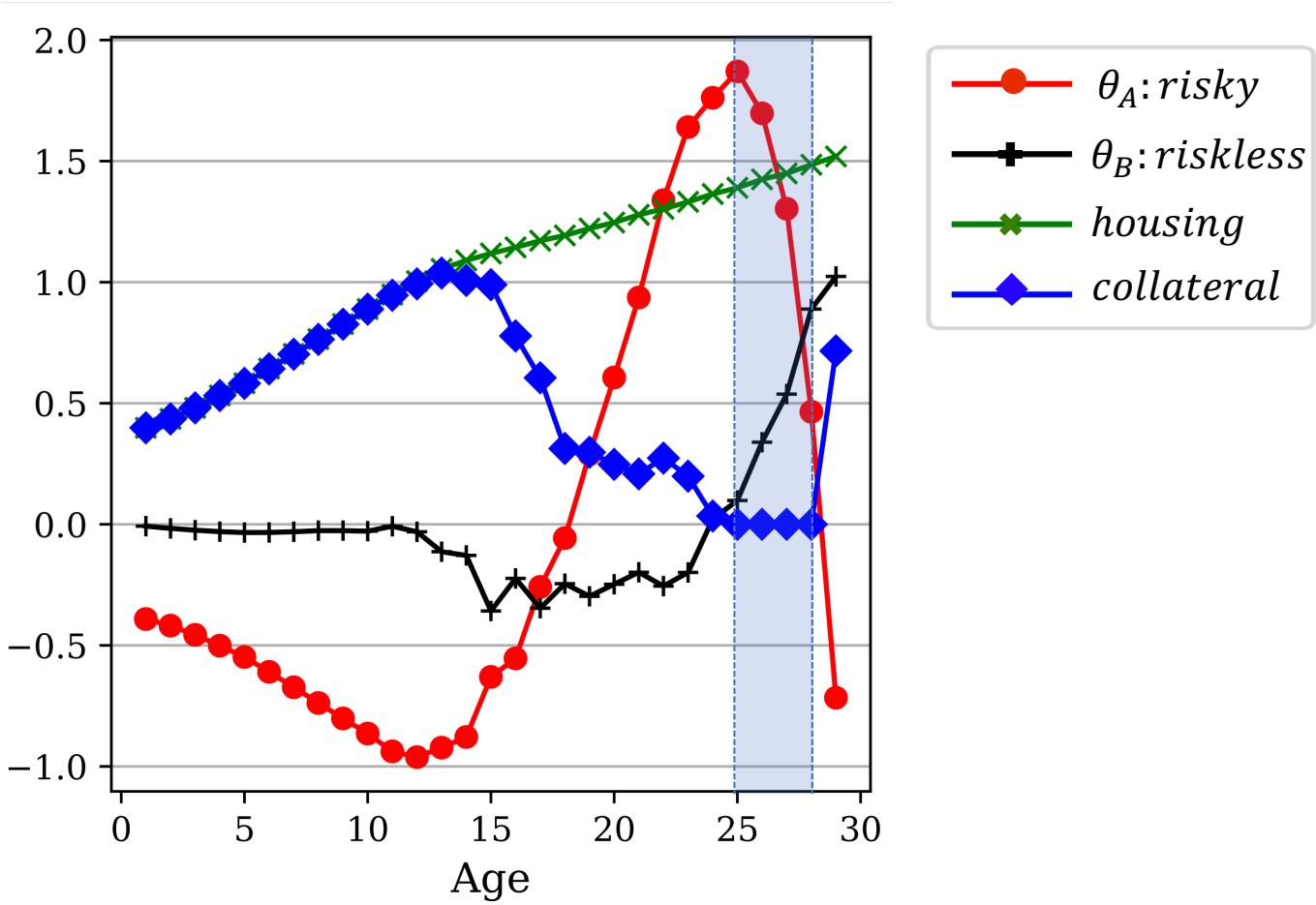
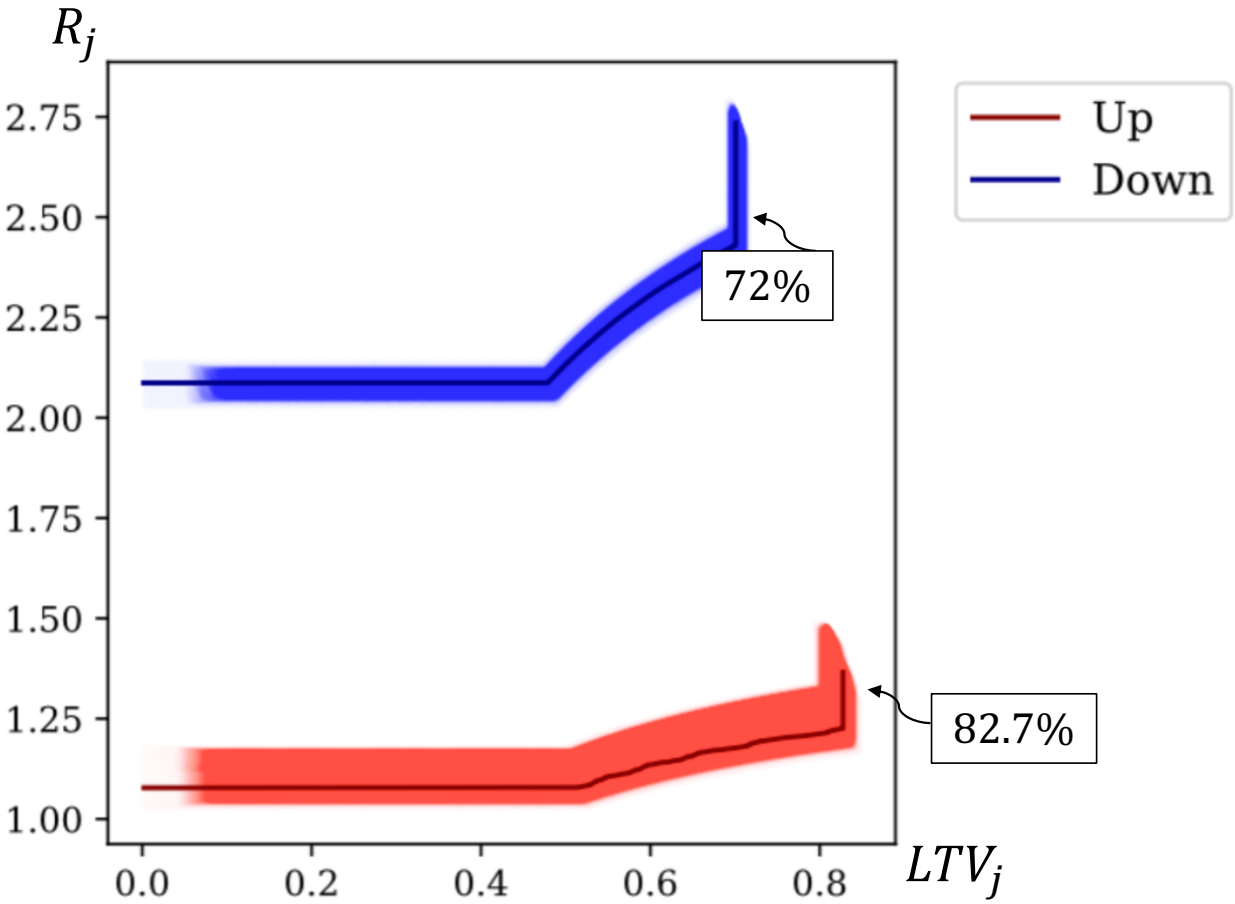
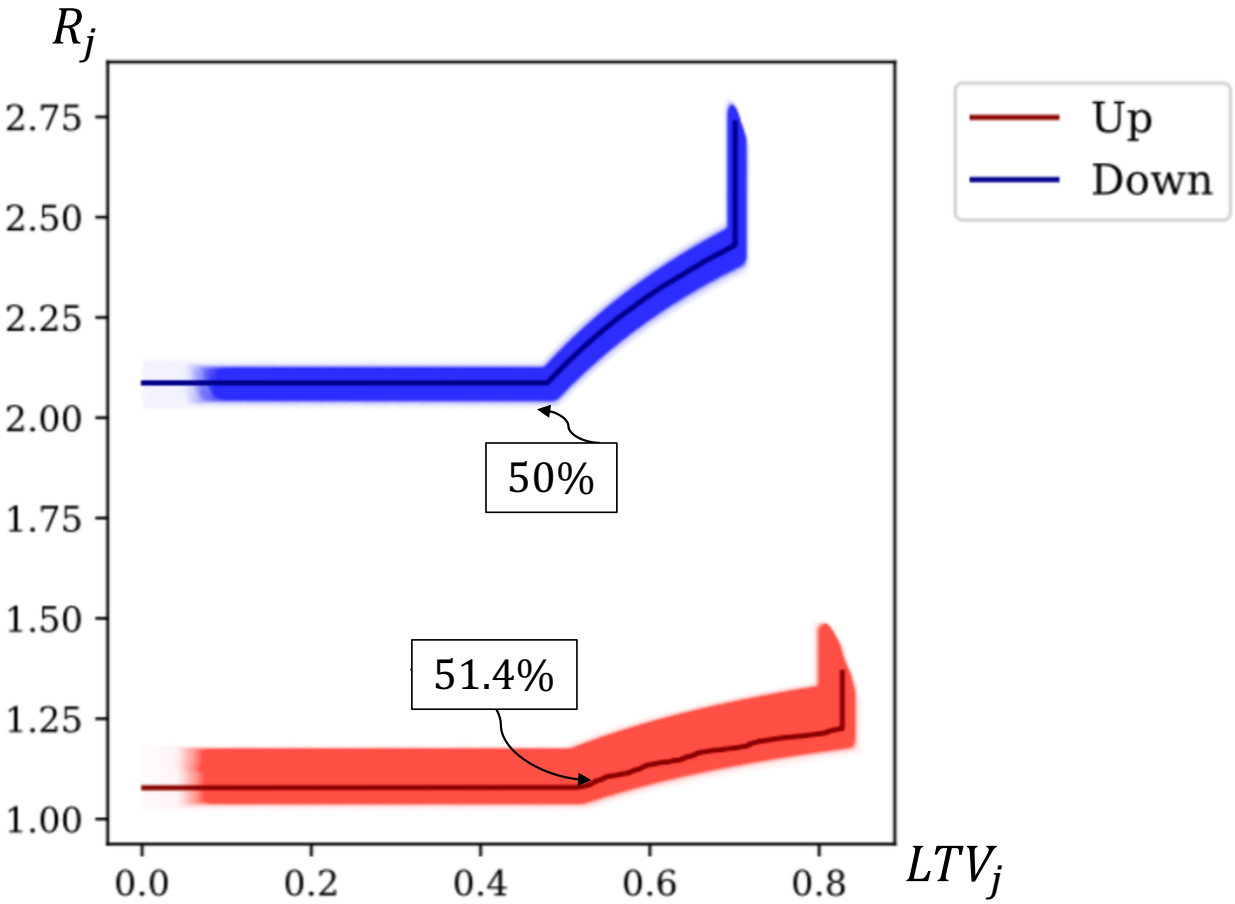


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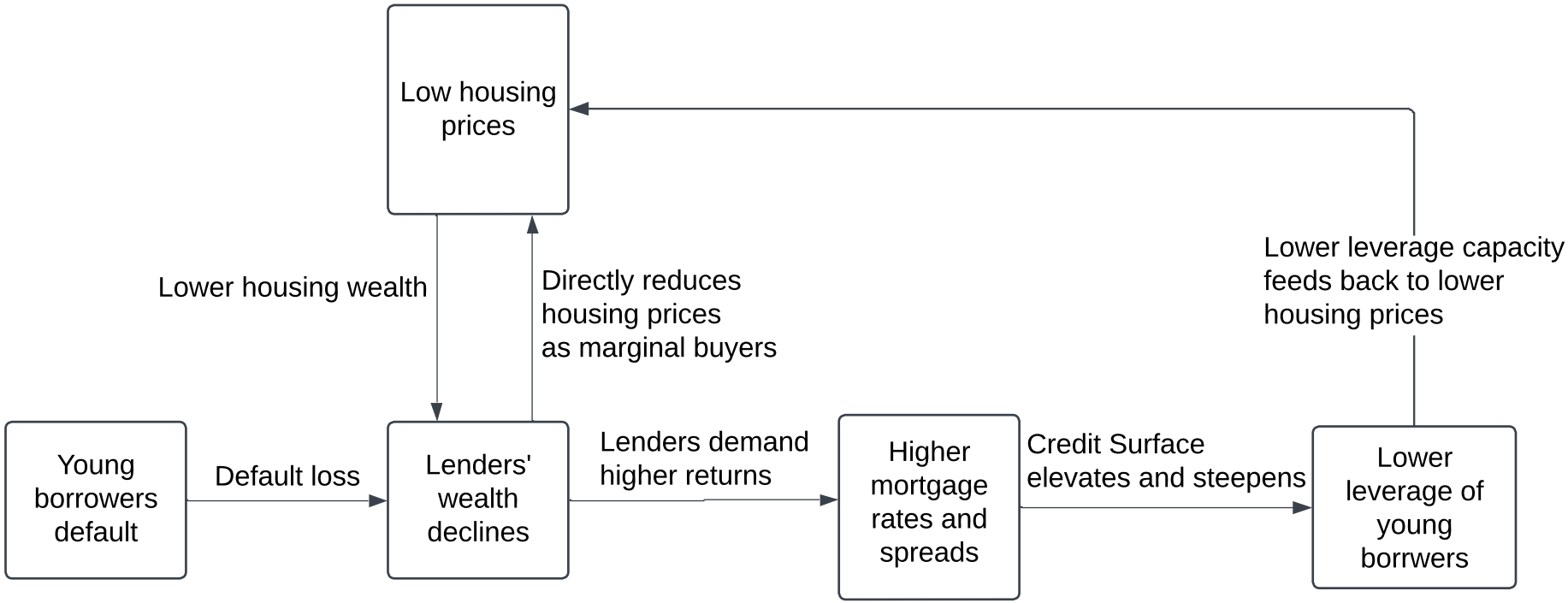
Credit Surface



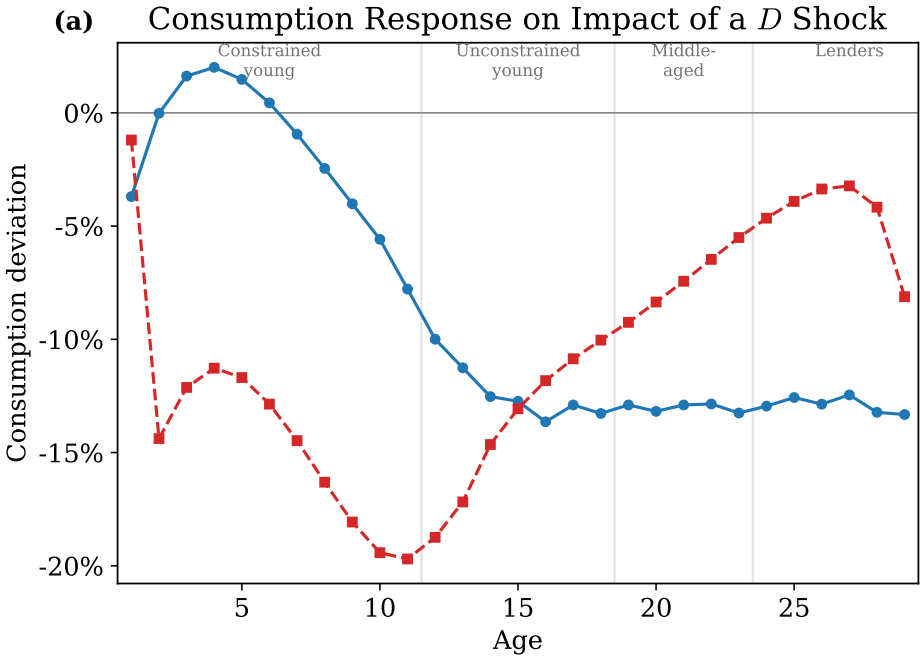
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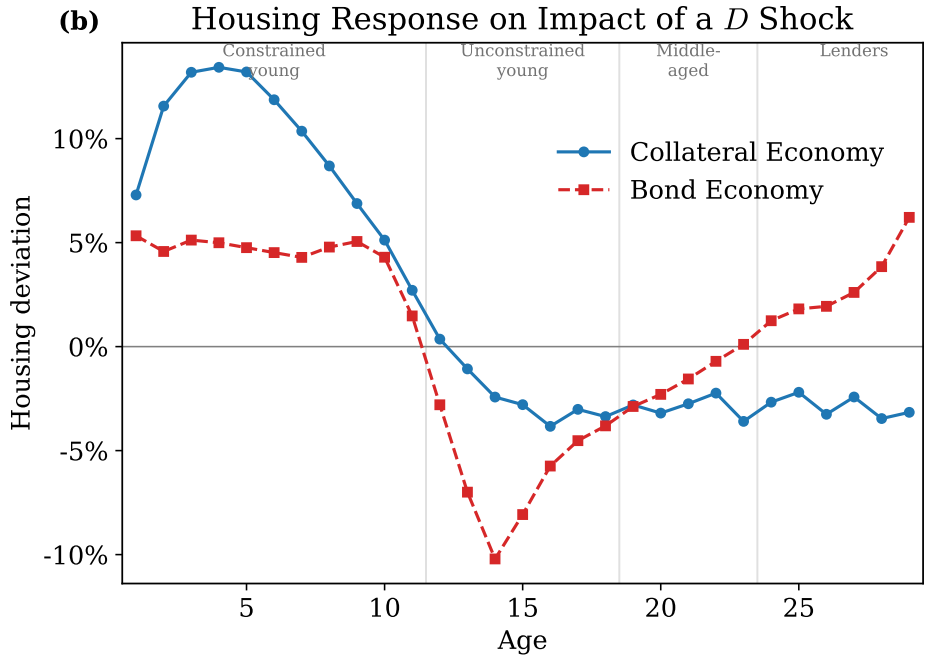


Amplification: Feedback Loop Between Leverage and Housing Prices



Consumption and Housing Response by Age





This paper

Goal: Rationalize two empirical facts in the U.S. housing market:

$$\text{house prices} \uparrow \quad \text{leverage} \uparrow \quad \text{mortgage spreads} \downarrow .$$

Approach: A quantitative OLG model with endogenous leverage pricing.

- Households borrow and lend through a menu of defaultable mortgage contracts.
- Each contract has a distinct LTV and interest rate: the **Credit Surface**.
- Leverage, spreads, and default risk are jointly determined in equilibrium.

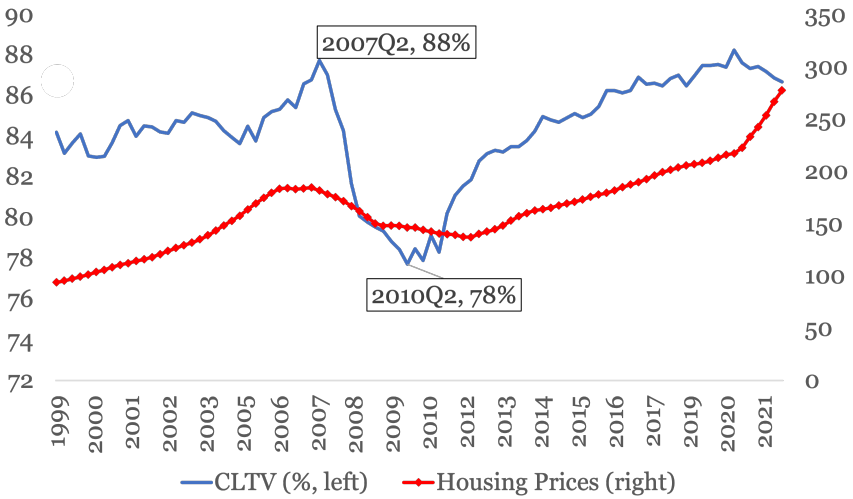
Main mechanism: Leverage amplifies housing-price volatility through two channels:

- **Wealth channel:** default losses reduce lenders' wealth and housing demand.
- **Leverage channel:** higher spreads reduce borrowers' leverage capacity.

Thank You!

Motivating Fact 1

- Leverage co-moves with housing prices (0.53)
- $\text{CLTV} \equiv \frac{\text{Total newly issued loan}}{\text{Appraised value of housing collateral}}$

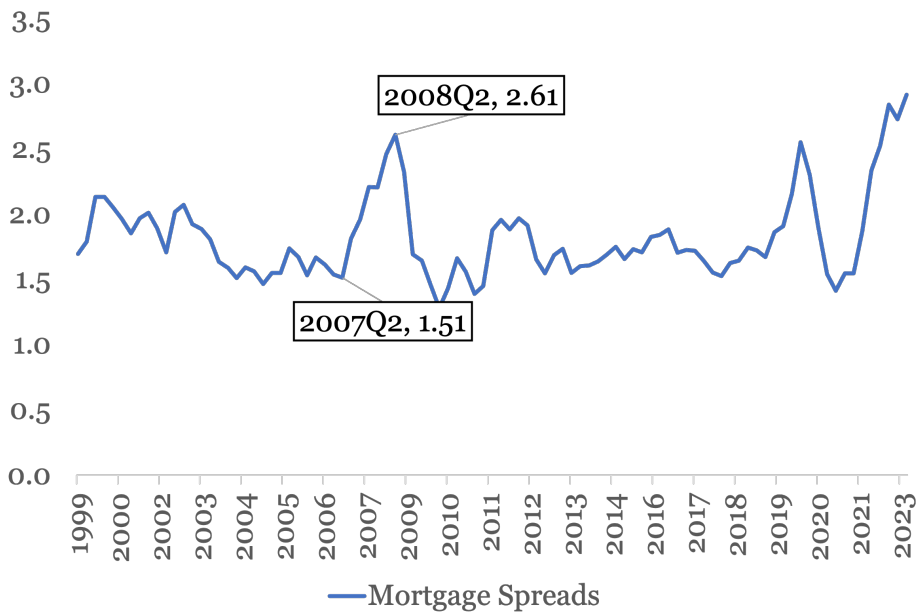


Data source: CLTV from Freddie Mac Single-Family Loan-Level Dataset, Case-Shiller housing price index from FRED, index Jan 2000=100.

Figure: Mean CLTV of First-time Homebuyers and Housing Prices

Motivating Fact 2

- Mortgage spreads move in opposite direction (-0.29)



Data source: FRED MORTGAGE30US, DGS10.

Mortgage spread: average 30-year fixed-rate mortgage - 5-year treasury bill yield

Two Challenges

- Conceptual Challenge
  - Infinitely many contracts, which ones are chosen in equilibrium?
- Computational Challenge
  - High degree of heterogeneity
  - Large aggregate shocks
  - Occasionally binding collateral constraint

Solutions

- Conceptual Challenge
  - Guess and verify
    - Guess only a finite number of contracts are traded, reducing KKT conditions to a finite number of equations
    - Verify if agents have incentives to deviate to trading contracts not included in the finite set
- Computational Challenge
  - Neural network

Parameters

Table: Parameters

Parameter	Description	Value
Demographics		
A	Life span	30
Aggregate uncertainty		
Γ	Transition matrix	$\begin{bmatrix} 0.86 & 0.14 \\ 0.77 & 0.23 \end{bmatrix}$
Preferences		
β	Discount factor	0.92
ρ	Risk aversion	5.5
α	Weight on housing	0.115
Age-endowment profiles		
$\{e^a(U)\}_{a=1}^{20}$	In state U	See Figure **??**
$\{e^a(D)\}_{a=1}^{20}$	In state D	See Figure **??**
Housing		
H	Housing supply	30

Aggregate Shocks Impact Housing Prices through Two Channels

$$q_t = \underbrace{\underbrace{\frac{Du_h^a(c_t^a, h_t^a)}{Du_c^a(c_t^a, h_t^a)}}_{\text{Rent}} + \underbrace{E_t\left[\beta \frac{Du_c^{a+1}(c_{t+1}^{a+1}, h_{t+1}^{a+1})}{Du_c^a(c_t^a, h_t^a)} q_{t+1}\right]}_{\text{PV}(q_{t+1})}}_{\text{Fundamental Value: Wealth Channel}} + \underbrace{\frac{\mu_{h,t}^a}{Du_c^a(c_t^a, h_t^a)}}_{\text{Collateral Value: Leverage Channel}}$$

Wealth Channel: Unconstrained Agents

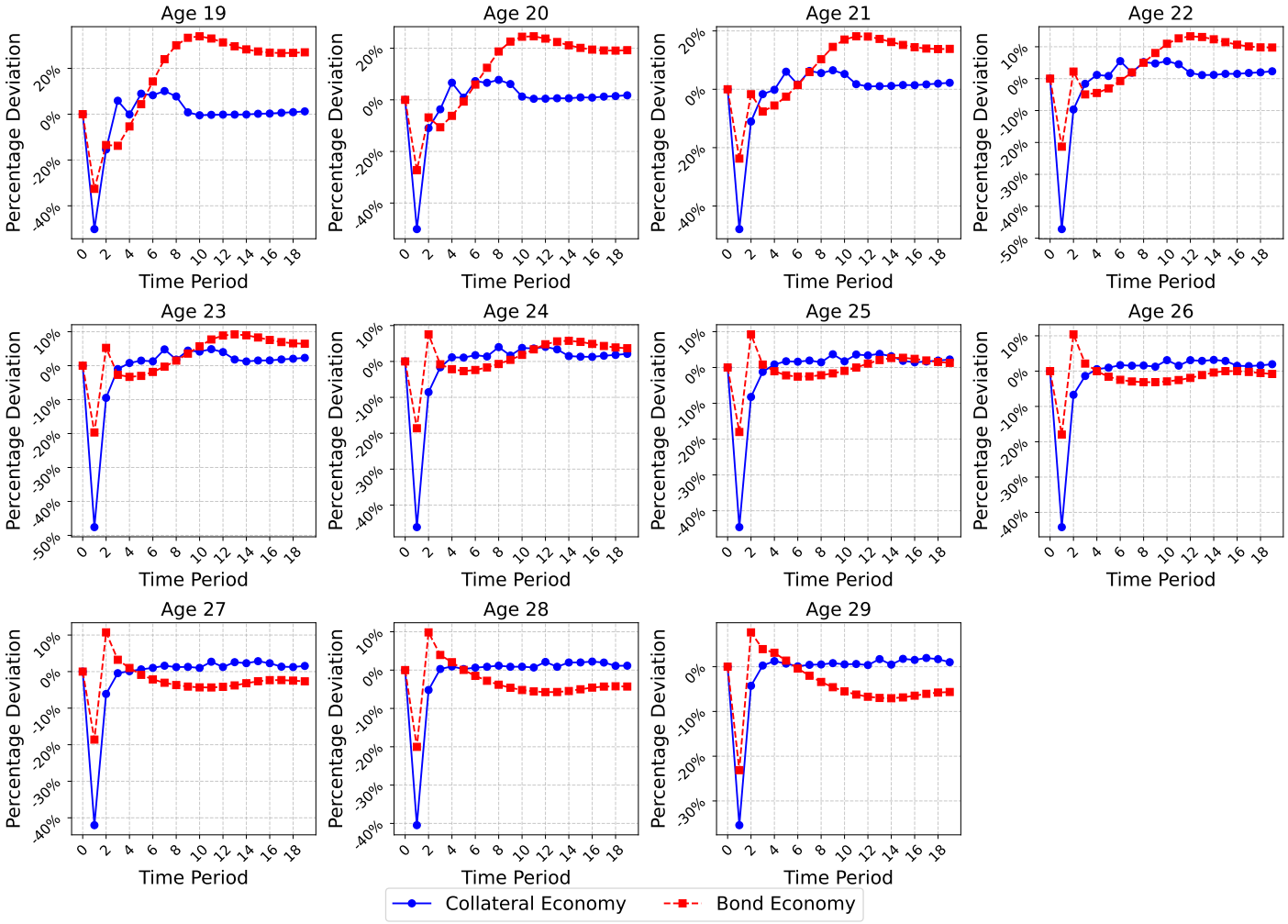


Figure: Percentage Deviations in Financial Wealth Under a D shock

Leverage Channel: Constrained Borrowers

